DiscoveryRecommendationSkill for Daily arXiv Paper Recommendation

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Abstract

DiscoveryRecommendationSkill is the recommendation component of the Daily arXiv Research Briefing Agent. It converts user interests, arXiv categories, date constraints, seed papers, and feedback into query plans, candidate pools, ranked recommendations, and local follow-up results. The Skill represents papers, topics, categories, seed preferences, retrieval runs, and feedback events as linked elements in an academic information network. It separates candidate generation from ranking, uses structured output contracts, and returns explicit statuses for success, empty, fallback, and error cases. The evaluation combines deterministic fixture tests with a frozen real-arXiv ranking study that compares deterministic ranking, semantic ranking, and BM25 under short keyword topics and longer intent-style topics.

1 Functionality

Daily arXiv discovery is a filtering task over a fast-changing stream of papers. A user rarely provides a fully specified retrieval query. The input more often contains a topic phrase, an arXiv category, a date range, seed papers, or feedback from previous recommendations. DiscoveryRecommendationSkill maps these signals into a ranked reading list before the synthesis Skill writes a daily briefing. In network terms, the Skill connects user interests to papers through categories, seed-paper relations, retrieval provenance, ranking rationales, and feedback events. The output consists of a QueryPlan, PaperMetadata records, Top-K Recommendation rows with scores and rationales, feedback rank movement, and FollowupSkill results over the local paper store.

The Skill serves as a public facade over smaller implementation units. SeedParsingSkill builds a SeedPreference from arXiv IDs, arXiv URLs, and title strings. QueryPlanningSkill expands a RetrievalQuery into required terms, optional terms, phrases, categories, and bounded query variants. ArxivRetrievalSkill queries the arXiv Atom API when cache reuse is insufficient, parses metadata, and writes retrieval runs to SQLite. TopicRankingSkill ranks candidates with lexical, phrase, category, recency, retrieval-source, seed, and feedback signals. SemanticSeedRankingSkill checks embedding readiness and compares seed and candidate representations. FeedbackRefinementSkill records paper-level like or dislike events and returns rank and score deltas. FollowupSkill first searches stored papers and fetches from arXiv only when local results are empty.

Figure 1 shows the Skill boundary and the main data relations. The design keeps inputs and outputs explicit because the downstream ResearchSynthesisSkill must identify whether a claim is supported by metadata, an abstract, ranking evidence, or later full-text extraction. This separation prevents weak retrieval evidence from being presented as paper-level analysis.

2 Method

The Skill first constructs a RetrievalQuery from the topic, category, date range, search mode, and retrieval budget. Query planning then controls how the request reaches arXiv. In deterministic mode,

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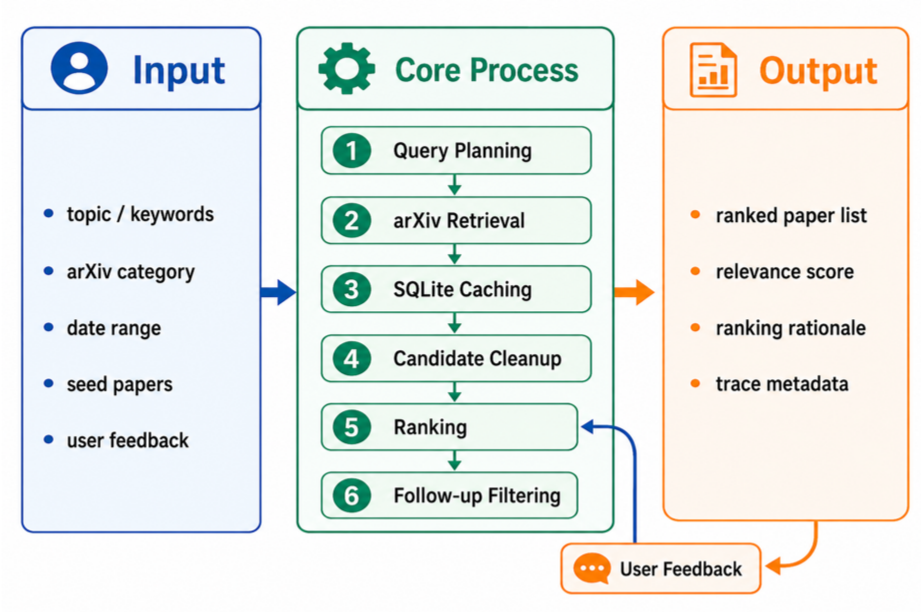

Figure 1: DiscoveryRecommendationSkill workflow. User interests are linked to query plans, retrieved arXiv metadata, ranked recommendations, feedback-adjusted rankings, and local-first follow-up results.

the planner preserves required topic terms and builds conservative query variants. In automatic or live-assisted mode, an LLM planner may propose broader terms, but invalid or divergent plans fall back to deterministic planning. This guard prevents recall-oriented expansion from changing the intent of the user.

Retrieval uses the arXiv Atom API and stores normalized metadata in SQLite. The cache contains papers and run-scoped retrieval metadata, including query variants and source positions, which makes candidate generation repeatable for the same effective query. Ranking is a separate stage. The deterministic ranker assigns each candidate a weighted score from term overlap, phrase matches, category agreement, recency, query-source evidence, seed preference text, and feedback events. Each recommendation includes a score breakdown and a rationale, which allows the user to inspect why a paper appears near the top.

The semantic path targets cases in which exact keywords are insufficient. It builds a seed representation from seed papers or topic-derived preference text, checks embedding configuration and seed quality, embeds seed and candidate metadata, and ranks candidates by cosine similarity with bounded lexical, category, and recency support. If semantic readiness fails, the Skill returns a structured error rather than treating deterministic ranking as semantic ranking. Feedback is paper-level: the Skill records each like or dislike event and reports the previous rank, new rank, score delta, and rank delta.

3 Related Work

The Skill follows standard ranked retrieval practice. Access to arXiv metadata uses the arXiv API (arXiv, 2026). The evaluation reports precision, recall, and mean reciprocal rank, which are common ranked retrieval measures (Manning et al., 2008). The ranking design is a hybrid content-based recommender because it combines topic text, category constraints, seed papers, retrieval provenance, and user feedback (Burke, 2002). The semantic path follows sentence embedding retrieval, where related documents are compared in a vector space rather than only through exact word overlap (Reimers and Gurevych, 2019). The broader Agent uses retrieval-augmented generation ideas, but this Skill stops at discovery and recommendation. Synthesis and full-paper explanation remain separate downstream responsibilities (Lewis et al., 2020).

Table 1: DiscoveryRecommendationSkill ranking results. The real-arXiv rows report macro averages over three topics and 150 frozen candidates.

Setting	Method	Precision@5	Recall@10	MRR
Fixture search	Planned retrieval and ranking	1.000	1.000	1.000
Short topic	Deterministic Agent	0.733	0.833	0.778
Short topic	Semantic Agent	0.733	0.722	1.000
Short topic	BM25	0.800	0.944	1.000
Intent topic	Deterministic Agent	0.600	0.556	0.667
Intent topic	Semantic Agent	0.733	0.722	0.750
Intent topic	BM25	0.467	0.611	0.833

4 Results

The evaluation has two layers. The first layer uses deterministic fixtures to verify Skill behavior under controlled inputs. Broad planned retrieval covers all expected relevant candidates in the search-quality fixture and ranks the three relevant papers in the top three, giving precision@3, recall@3, and MRR of 1.0. Rationale coverage is also 1.0. The feedback test reports rank and score movement after user feedback, and the local follow-up test returns stored matches without an arXiv request when SQLite already contains relevant papers.

The second layer uses a frozen real-arXiv ranking set with three topics, 50 recent candidates per topic, and binary human relevance labels. A paper is labeled relevant when the title or abstract matches the intended research need of the topic rather than only repeating an isolated keyword. Table 1 reports macro averages. Short keyword topics favor lexical retrieval because many relevant papers share query words with the topic. In this setting, BM25 gives the strongest precision@5 and recall@10. The deterministic Agent ranker remains competitive but has lower MRR because the first relevant paper is not always ranked first. The semantic Agent with a live embedding model improves MRR to 1.0 but does not exceed the recall of BM25.

To test whether semantic ranking helps when a query expresses intent rather than exact keywords, the frozen evaluation is rerun with longer paraphrased topics. For example, retrieval-augmented generation is expressed as question answering that consults external documents, grounds claims in evidence, attaches citations, and avoids unsupported statements. In this setting, Semantic Agent reaches macro precision@5 of 0.733 and recall@10 of 0.722, while BM25 drops to 0.467 and 0.611. This result supports the intended semantic use case, where user interests are closer to research intent than to paper-title keywords.

5 Analysis

The results show that the strongest behavior of the Skill depends on the input regime. When the topic is a short keyword phrase such as retrieval augmented generation or vision-language robotic manipulation, BM25 is a strong baseline because words in titles and abstracts overlap directly with the query. This outcome is not a weakness of the evaluation; it is a useful sanity check. A recommender should not claim semantic superiority when lexical matching already solves the task well.

The intent-topic experiment gives a clearer view of semantic ranking. In that setting, exact query words no longer cover all relevant papers, and the live embedding path improves both precision@5 and recall@10 over BM25. The improvement is most visible for evidence-grounded generation and open-vocabulary robotic manipulation. The LLM/tool-agent topic remains harder because the label space mixes embodied agents, software agents, benchmark agents, and multi-agent reinforcement learning. A stronger version should use seed papers rather than topic text alone, because seed abstracts define the target subfield more precisely than a long natural-language query.

The ablations also identify practical failure modes. Query expansion improves coverage only when planner outputs preserve required terms. Semantic ranking depends on embedding configuration and seed quality, so readiness checks and fail-closed behavior are necessary. Feedback refinement reports rank movement, but a few events cannot replace a well-labeled preference profile. Local-first follow-up reduces latency and arXiv calls, but it depends on sufficient local papers. Future work should add more topics, seed-paper labels, annotation checks, and longitudinal feedback logs.

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